

$$\lambda = \lambda_0 \quad y > 0$$

$$\lambda = -\lambda_0 \quad y < 0$$

$$b) \quad V_P = k \int \frac{dq'}{|r-r'|}$$

$$dq' = \lambda \cdot dy$$

$$r = (x, 0, 0)$$

$$r' = (0, y, 0)$$

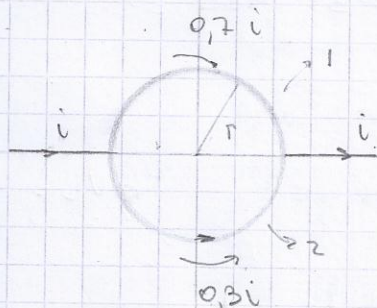
$$\vec{E} \cdot d\vec{r} = \vec{E} \cdot d\vec{r} \cdot \cos 90^\circ = 0$$

$$V_P = \frac{k}{2L} \int_{-L}^L \frac{\lambda \cdot dy}{y}$$

$$V_P = -\frac{k}{2L} \lambda \cdot \ln\left(\frac{L}{L}\right) = 0$$

Esse é o mesmo potencial $\Rightarrow 0$.

y



$$R = 1 \text{ m}$$

$$i = 100 \cdot 10^{-3} \text{ A}$$

$$2) \quad \vec{B}_{(0,0,0)}$$

$$\vec{B}_T = \vec{B}_2 - \vec{B}_1$$

$$B_T = \frac{\mu_0 \cdot 0.3i}{4r}$$

$$B = \frac{\mu_0 i}{4\pi r} \int \frac{d\vec{r}' \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

$$r = (0, 0, 0)$$

$$r = (R \cos \theta, R \sin \theta, 0)$$

$$r' = (-R \sin \theta, R \cos \theta, 0) d\theta$$

$$r = (-R \cos \theta, -R \sin \theta, 0) \quad x = (0, 0, R^2)$$

$$r^2 = R^2$$

$$B_1 = \frac{\mu_0 i}{4\pi R} \int_{\pi}^0 d\theta = \frac{\mu_0 \cdot 0.7i}{4\pi R} \cdot \pi (-1) = \frac{\mu_0 \cdot 0.7i}{4\pi R} (-1)$$

$$B_2 = \frac{\mu_0 \cdot 0.3i}{4\pi R} (1) \Rightarrow B_T = \frac{\mu_0 i}{4\pi R} (0.3 - 0.7)$$

$$B_T = 1.25 \cdot 10^{-8} \text{ T } (-1)$$